Test / raw: Cantera versus NN | paired short and long

Gamma chosen on long-run validation by completion count, then temperature NRMSE. Each pair uses the SAME gamma and common completed test ICs.
NRMSE = RMSE(Cantera, NN) / RMS(Cantera); dimensionless ratio, median across the listed ICs. Lower is better. Green/red: long improved/worsened.

	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2
ANN, g=0.1 | short	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ANN, g=0.1 | long	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
P-only, g=0.2 | short	0.05057	0.1949	0.3226	0.6239	0.3374	0.4099	0.1515	0.6485	0.8358	7.374e-17
P-only, g=0.2 | long	0.03343	0.1256	0.2608	0.3464	0.3456	0.2174	0.04444	0.7907	0.8048	7.374e-17
ANN + P-only, g=0.2 | short	0.0463	0.1758	0.2418	0.8481	0.5589	0.5166	0.2856	0.8109	0.8214	4.482e-16
ANN + P-only, g=0.2 | long	0.02843	0.12	0.2832	0.3985	0.2723	0.2132	0.08543	0.5102	0.4091	4.482e-16
T/P, g=0.3 | short	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
T/P, g=0.3 | long	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ANN + T/P, g=0.3 | short	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ANN + T/P, g=0.3 | long	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
T/P/phi, g=0.2 | short	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
T/P/phi, g=0.2 | long	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ANN + T/P/phi, g=0.2 | short	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ANN + T/P/phi, g=0.2 | long	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

ANN: common n=0, complete short=3/18, long=6/18; ICs=[]
P-only: common n=4, complete short=13/18, long=5/18; ICs=[39, 44, 45, 73]
ANN + P-only: common n=3, complete short=13/18, long=4/18; ICs=[39, 44, 45]
T/P: common n=0, complete short=0/18, long=0/18; ICs=[]
ANN + T/P: common n=0, complete short=0/18, long=2/18; ICs=[]
T/P/phi: common n=0, complete short=1/18, long=3/18; ICs=[]
ANN + T/P/phi: common n=0, complete short=1/18, long=1/18; ICs=[]

Short: 10,000 updates. Long: scalar training-MSE plateau or 150 s of cumulative new active work/model; plateau 0/240.
Gamma=0.1 ANN/T-P origins were rebuilt because saved Adam states were unavailable; bootstrap counts within 150 s. Other 220 origins preserve exact Adam/RNG.
Incomplete profiles have no full-horizon score. Pairwise cohorts may differ between teams; these rows do not establish an overall architecture ranking or all-species optimum.